

Coupling Interpretable Allocation, Model-Predictive Control, and Tail-Aware Forecasting for Live Portfolio Management

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Outline

Live portfolio management couples three problems: allocation, the timing of rebalancing, and evaluation beyond the historical record.

1 Allocation — What to hold. Maximize utility of a score γ for each stock, within trading limits.

2 Rebalancing — When to trade. Re-plan at the 21-day horizon, a band exit, a drawdown stop, or a basket change; real-data backtest, 2025–26.

3 Market generator — Evaluation. A backtest is one historical path; JumpHMM generates typical and stressed futures.

We want to build evolving portfolios that maximize investor satisfaction as preferences shift with the market.

Maximize utility subject to trading limits (convex cone)

Each rebalance chooses n_i shares of each name i to solve:

$$\max_{n \geq 0} \; U(n;\gamma)$$ utility U (γ_i: score of name i, next slide)

s.t. $\sum_i n_i S_i \leq W$ budget W (S_i: price of name i)

$w_i \leq \bar{w}$ single-name cap $\bar{w}$ (w_i: weight of name i)

$\sqrt{w^\top \Sigma_g \, w} \leq \sigma_{\text{SPY}}$ volatility cap σ_{SPY} (Σ_g: growth-rate covariance)

$\bar{c} \, \|n - n^{\text{prev}}\|_1 \leq B$ turnover budget B ($\bar{c}$: half spread per share)

$\beta^\top w \leq \bar{\beta}$ β cap $\bar{\beta}$, hedge on (β: beta to SPY)

	Cobb–Douglas	CES on shares	CES on weights	Stone–Geary
$U(n;\gamma)$, over $\gamma_i > 0$	$\sum \gamma_i \ln n_i$	$(\sum \gamma_i n_i^\rho)^{1/\rho}$	$(\sum \gamma_i w_i^\rho)^{1/\rho}$	$\sum \gamma_i \ln(n_i - b_i)$
budget alone	$w_i \propto \gamma_i$	$n_i \propto (\gamma_i / S_i)^\eta$	$w_i \propto \gamma_i^\eta$	$(n_i - b_i)\, S_i \propto \gamma_i$

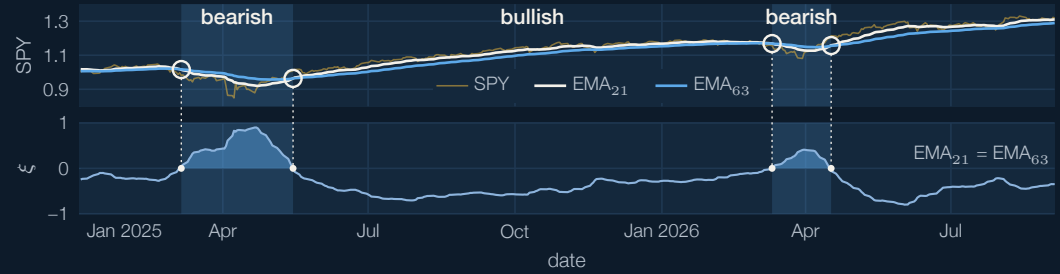
$\rho = 1 - 1/\eta$, $\eta = 0.5$; $b_i S_i = 1\%$ of W. Also compared: minimum variance, maximum Sharpe, 130/30 long-short.

The score γ: expected growth, tilted by the market

$$\gamma_i = \tanh\left(\frac{\alpha_i + \beta_i\,\tilde{g}_M}{\max(|\beta_i|,\,0.1)^{\xi}}\right)$$

expected growth (α_i, β_i: daily fit of name i to SPY; $\tilde{g}_M$: SPY's growth rate, 252-day EMA)

β penalty (ξ: market state, >0 when $\text{EMA}_{21} < \text{EMA}_{63}$)



Weighting γ enters $U(n;\gamma)$: names with $\gamma_i > 0$ are held long, and 130/30 shorts $\gamma_i < 0$.

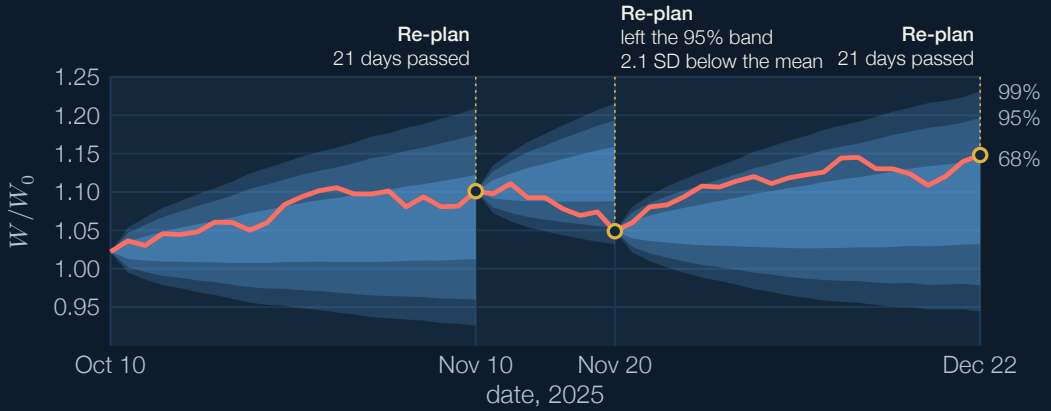
Selection the K highest γ overall, or the m highest in each sector.

$\xi = -G\,(\text{EMA}_{21}/\text{EMA}_{63} - 1)$ from SPY's 21- and 63-day exponential averages; $|\xi| \le 1$ on 99% of 2014–2024 days.

Markets drift away from any plan, so when should the portfolio stop and re-plan?

Rebalancing triggers: horizon, forecast band, drawdown

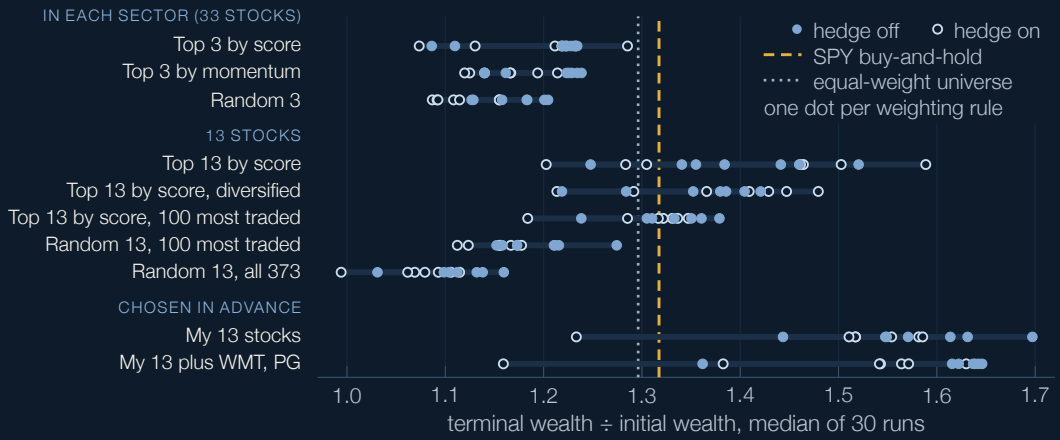
Forecast: multi-asset GBM, re-fit at each re-plan, 21 days ahead. Plant: real prices.



My 13 stocks, Cobb–Douglas, hedge off; over Jan 2025 – Sep 2026, 27 re-plans.

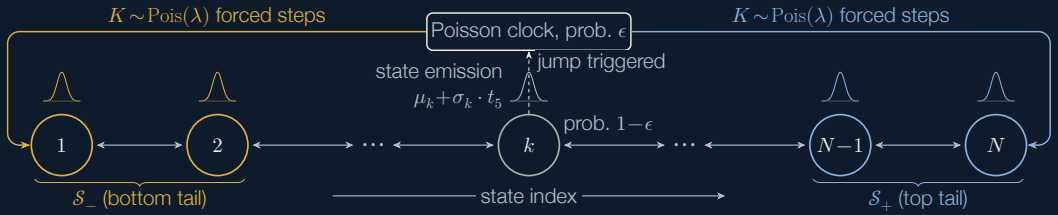
Stock selection matters more than the weighting rule

Real-data backtest, Jan 2025 – Sep 2026, 373 names. Each row is a selection rule.



What if 2025–26 was just a good run? Let's test our approaches on fat-tailed, multi-asset synthetic markets.

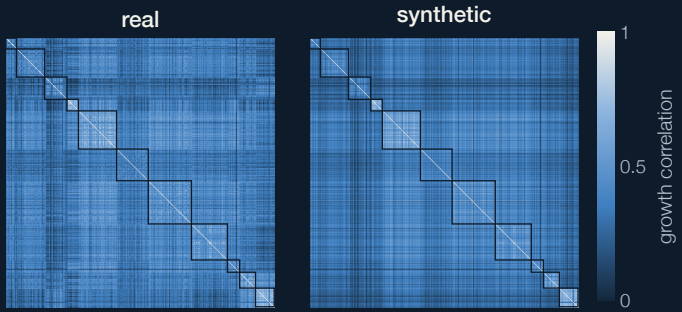
JumpHMM: a market generator with fat tails



The controller forecasts with GBM; JumpHMM generates the test markets. Alswaidan and Varner, arXiv:2603.10202.

Synthetic names move together like the real ones

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$



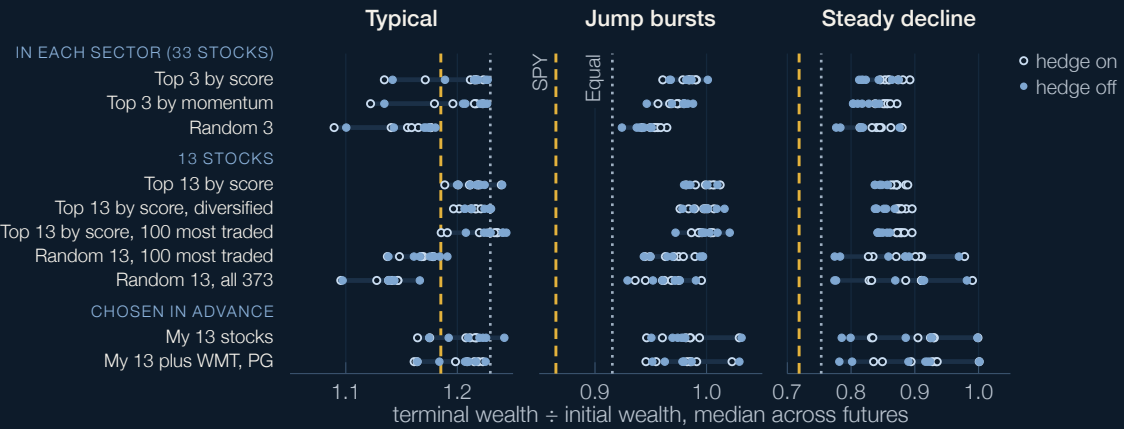
mean pairwise correlation

	within a sector	across sectors
real	0.47	0.35
synthetic	0.45	0.31

Real: 373 names, 2014–2024. Synthetic: one simulated path, same dates. Names sorted by sector; sectors outlined.

Simulated sell-offs: every median ends above SPY

Of 10,000 futures: 600 typical (random), plus the 200 worst SPY drops of each kind.



Jump bursts: months of extreme up and down days. Steady decline: no such stretch, a deeper net fall.

Key takeaways

- **Allocation: stock selection mattered more than the weighting rule.** With the hedge off, seven weighting rules on one 33-name basket ended at 1.09–1.23 times initial wealth; different baskets ended at 1.03–1.70.
- **Rebalancing: re-plans came at the horizon, a band exit, a drawdown stop, or a basket change.** About 27 re-plans in 419 trading days, a third of them before the 21-day horizon.
- **Market generator: the results held up beyond 2025–26.** In 600 typical simulated futures, 13 stocks picked by γ beat 13 random stocks 62–71% of the time, and in simulated sell-offs every median still ended above SPY. In a slow bear market, random picks did as well as γ for most weighting rules.

A paper-trading version of this system is running now; its results are not part of this talk.

Acknowledgements

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helpful discussions

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an early version of the Cobb–Douglas allocation and the bandit picker

We also thank Alpaca Markets, and Polygon.io for the original daily price data.

Paper

arXiv:2611.03022

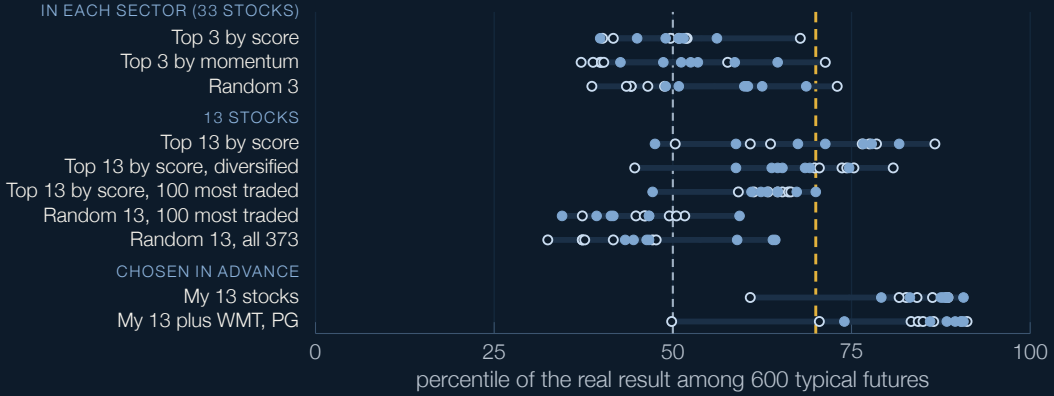
Code

github.com/varnerlab/INFORMS-Annual-Meeting-2026-Varnerlab

Questions: jdv27@cornell.edu

Real results as percentiles of typical futures

Each real result among 600 typical JumpHMM futures. Right of 50 beats the median future.

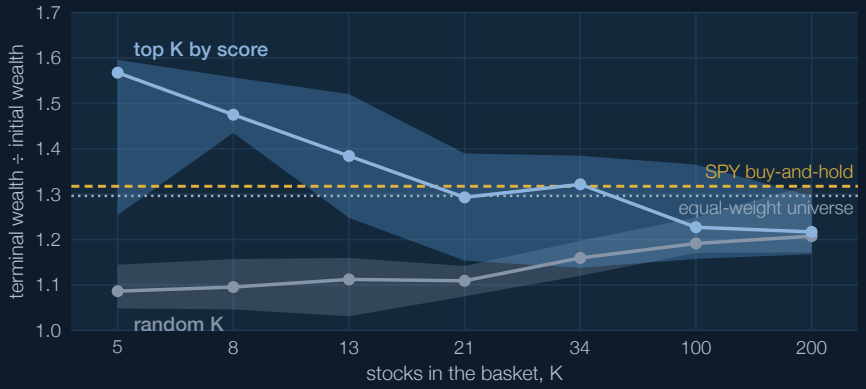


● hedge off ○ hedge on – – · SPY buy-and-hold one dot per weighting rule

Real results use the simulated runs' seeds: 5660–5664 for random and bandit pickers, 5660 for the rest.

Top K by score against random K, 5 to 200 stocks

The K highest γ against K random stocks. Line: median over the weighting rules, each the median of 30 runs; band: lowest to highest rule (at 5 stocks, the low edge is 130/30).



Real-data backtest, Jan 2025 – Sep 2026, hedge off. Stone–Geary is left out at 100 and 200 stocks.

Terminal wealth by picker and utility: hedge off

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
IN EACH SECTOR (33 STOCKS)							
Top 3 by score	1.230	1.219	1.221	1.234	1.086	1.227	1.110
Top 3 by momentum	1.226	1.234	1.223	1.239	1.162	1.229	1.140
Random 3	1.205	1.183	1.201	1.183	1.128	1.158	1.127
13 STOCKS							
Top 13 by score	1.520	1.384	1.460	1.441	1.355	1.341	1.248
Top 13 by score, diversified	1.352	1.386	1.421	1.380	1.284	1.404	1.219
Top 13 by score, 100 most traded	1.333	1.310	1.379	1.350	1.238	1.361	1.305
Random 13, 100 most traded	1.216	1.173	1.211	1.212	1.158	1.274	1.152
Random 13, all 373	1.098	1.132	1.138	1.112	1.104	1.160	1.031
CHOSEN IN ADVANCE							
My 13 stocks	1.548	1.571	1.614	1.631	1.443	1.697	1.549
My 13 plus WMT, PG	1.616	1.646	1.637	1.639	1.362	1.642	1.622
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

CD Cobb–Douglas · CES / CESw CES on shares / weights · SG Stone–Geary · MinVar minimum variance · Tangent tangency · LS 130/30 long-short.

Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.

Terminal wealth by picker and utility: hedge on

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
IN EACH SECTOR (33 STOCKS)							
Top 3 by score	1.212	1.232	1.223	1.218	1.073	1.285	1.130
Top 3 by momentum	1.125	1.140	1.167	1.194	1.166	1.214	1.120
Random 3	1.108	1.087	1.114	1.155	1.155	1.092	1.092
13 STOCKS							
Top 13 by score	1.503	1.305	1.464	1.461	1.203	1.589	1.283
Top 13 by score, diversified	1.409	1.429	1.479	1.447	1.213	1.366	1.292
Top 13 by score, 100 most traded	1.322	1.285	1.347	1.331	1.184	1.317	1.336
Random 13, 100 most traded	1.167	1.156	1.174	1.178	1.123	1.155	1.112
Random 13, all 373	1.069	1.106	1.093	1.115	1.079	1.062	0.994
CHOSEN IN ADVANCE							
My 13 stocks	1.554	1.517	1.517	1.511	1.233	1.582	1.586
My 13 plus WMT, PG	1.564	1.571	1.543	1.542	1.159	1.383	1.630
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

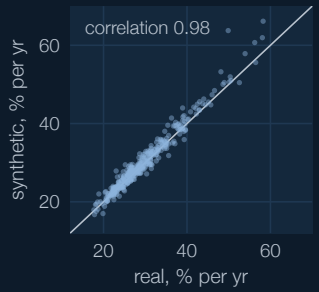
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Terminal wealth ÷ initial wealth, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.

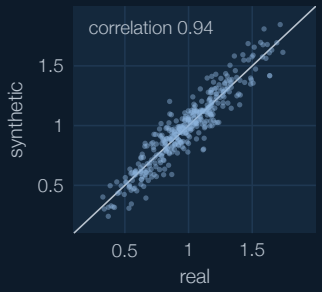
Each synthetic name resembles its real counterpart

$$g_i(t) = \underbrace{\alpha_i}_{\text{trend}} + \underbrace{\beta_i(t)\, g_m(t)}_{\text{market: 1 JumpHMM}} + \underbrace{\delta_i\, f_s(t)}_{\text{sector: 11 JumpHMMs}} + \underbrace{s_i\, \tilde{\varepsilon}_i(t)}_{\text{own: 373 JumpHMMs}}$$

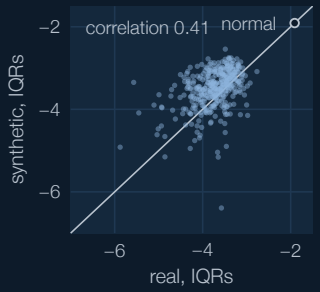
volatility



β against SPY



worst day in 200



Each dot is a name, real 2014–2024 against one simulated path. IQR: interquartile range of daily growth.